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C02
AT-1
Operating Systems
CSA-0401

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1) CPU Scheduling - SJF (Non-preemptive)

Execution Order: $P4 \rightarrow P2 \rightarrow P3 \rightarrow P1$

a) Waiting Time & Turnaround time:

Waiting Time = Start Time - Arrival Time

Turnaround Time = Waiting Time + Burst Time

Process	Burst Time	Waiting Time	Turnaround Time
P4	2	0	2
P2	4	2	6
P3	6	6	12
P1	10	12	22

b) Average Waiting Time:

$$= \frac{0+2+6+12}{4} \Rightarrow \frac{20}{4} \Rightarrow 5 \text{ ms}$$

Average Turnaround Time:

$$= \frac{2+6+12+22}{4} = \frac{42}{4} = 10.5 \text{ ms.}$$

c) Interpretation:

FCFS order:

$P1 \rightarrow P2 \rightarrow P3 \rightarrow P4$

Waiting Time:

$$P1 = 0$$

$$P2 = 10$$

$$P3 = 14$$

$$P4 = 20$$

Avg. Waiting Time:

$$= \frac{0 + 10 + 14 + 20}{4} = 11 \text{ ms.}$$

Algorithm	Avg. Waiting Time	Avg. Turnaround
FCFS	11 ms	16.5 ms
SJF	5 ms	10.5 ms

2) Round Robin Scheduling:

Time Quantum = 4 ms.

a) Gantt chart:

$|P1|P2|P3|P4|P1|P2|P3|P4|P1|P3|P1|P3|P1$

b) Waiting Time & Turnaround Time:

$$P_2 = 21$$

$$P_4 = 29$$

$$P_3 = 42$$

$$P_1 = 46$$

Turnaround Time = Completion Time.

Waiting Time = Turnaround - Burst

Process	Burst	Completion	Turnaround	Waiting
P1	20	46	46	26
P2	5	21	21	16
P3	13	42	42	29
P4	8	29	29	21

Average Waiting Time:

$$= \frac{26 + 16 + 29 + 21}{4} = \frac{92}{4} = 23 \text{ ms}$$

Average Turnaround Time:

$$= \frac{46 + 21 + 42 + 29}{4} = \frac{138}{4} = 34.5 \text{ ms.}$$

c) Effect of Quantum = 10 ms.

Advantages:

- * Fewer context switches.
- * Less CPU overhead

* Better CPU efficiency.

Disadvantages:

- * Longer waiting time for short ^{processes}
- * slower response time.
- * Behaves more like FCFS.

3) Banker's Algorithm

Total Resources:

$$A = 10$$

$$B = 5$$

$$C = 7$$

Allocated Resources:

$$A = 0 + 2 + 3 + 2 + 0 = 7$$

$$B = 1 + 0 + 0 + 1 + 0 = 2$$

$$C = 0 + 0 + 2 + 1 + 2 = 5$$

Available:

$$A = 10 - 7 = 3$$

$$B = 5 - 2 = 3$$

$$C = 7 - 5 = 2$$

$$\text{Available} = (3, 3, 2)$$

a) Need Matrix:

Need = maximum - Allocation.

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Process	A	B	C
P1	7	4	3
P2	1	2	2
P3	6	0	0
P4	2	1	1
P5	5	3	1

b) Safe state check:

$$\text{Available} = (3, 3, 2)$$

(i) $P2 = \text{Need } (1, 2, 2)$

$$\begin{aligned} \text{Possible} &= (3, 3, 2) + (1, 2, 2) \\ &= (5, 3, 2) \end{aligned}$$

(ii) $P4 \text{ need} = (2, 1, 1)$

$$\begin{aligned} \text{Available} &= (5, 3, 2) + (2, 1, 1) \\ &= (7, 4, 3) \end{aligned}$$

(iii) $P5 \text{ Need} = (5, 3, 1)$

$$\begin{aligned} \text{Available} &= (7, 4, 3) + (5, 3, 1) \\ &= (12, 7, 4) \end{aligned}$$

(iv) $P1 \text{ Need} = (7, 4, 3)$

$$\text{Available} = (12, 7, 4) + (7, 4, 3)$$

$$= (7, 5, 5)$$

Step 5

(4)

$$P3 \text{ Need} = (6, 0, 0)$$

Available:

$$(7, 5, 5) + (3, 0, 2)$$

$$= (10, 5, 7)$$

c) Safe Sequence:

$$P2 \rightarrow P4 \rightarrow P5 \rightarrow P1 \rightarrow P3$$

Therefore, SAFE STATE.

4) Resource Allocation Graph:

a) Text representation:

Assignments:

$$R1 \rightarrow P1$$

$$R2 \rightarrow P2$$

$$R3 \rightarrow P3$$

Requests:

$$P1 \rightarrow R2$$

$$P2 \rightarrow R3$$

$$P3 \rightarrow R1$$

Graph:

$R_1 \rightarrow P_1 \rightarrow P_2 \rightarrow R_3 \rightarrow P_3 \rightarrow R_1$

b) Deadlock:

P_1 waits for R_2

↓

P_2 waits for R_3

↓

P_3 waits for R_1

↓

back to P_1

Hence, deadlock exists;

c) Recommended Strategy:

Recovery:

Terminate one process or preempt R_3 .

This breaks the circular wait -

Justification:

Recovery is suitable because the deadlock has already occurred. The operating system can terminate one process or reclaim resources to normal execution.

5) CPU Utilization:

$$U = 1 - p^n$$

$\left\{ \begin{array}{l} p = \text{probability waiting for I/O} \\ n = \text{degree of multiprogramming} \end{array} \right.$

given: $p = 0.8$

a) Degree = 4

$$\begin{aligned} U &= 1 - (0.8)^4 \\ &= 1 - 0.4096 \\ &= 0.5904 \\ &\Rightarrow 59.04\% \end{aligned}$$

b) Degree = 6

$$\begin{aligned} U &= 1 - (0.8)^6 \\ &= 1 - 0.262144 \\ &= 0.737856 \\ &\Rightarrow 73.79\% \end{aligned}$$

c) Interpretation:

Degree of Multiprogramming: CPU utilization

4

59.04%

6

73.79%